Allergic Reaction / Anaphylaxis

AM-01 | Adult Medical

Quick flow

01 Suspected Allergic Reaction [EMT / AEMT / Paramedic]

Remove trigger when safe • assess airway, breathing, circulation, skin, GI symptoms, and perfusion

02 Anaphylaxis? [EMT / AEMT / Paramedic]

Respiratory compromise • hypotension/poor perfusion • or 2 or more body systems

03 IM Epinephrine Now [EMT / AEMT / Paramedic]

1 mg/mL: 0.5 mg IM lateral thigh • 0.3 mg autoinjector acceptable • repeat every 5 min

04 Position + Support Airway [EMT / AEMT / Paramedic]

Supine when tolerated • oxygen/suction/BVM • albuterol for persistent bronchospasm • early advanced airway

05 Hypotension / Poor Perfusion? [AEMT / Paramedic]

IV/IO isotonic crystalloid 500 mL aliquots • repeat with reassessment • maximum 2 L

06 Refractory Shock? [Paramedic / Medical Control]

Continue IM epinephrine • epinephrine infusion 2-10 mcg/min • Medical Control required

07 Bradykinin Angioedema? [Paramedic / Medical Control]

Early difficult-airway plan • consider TXA 1 g IV/IO over 10 min with Medical Control

08 Monitor + Transport [EMT / AEMT / Paramedic]

Continuous cardiac/SpO_2/BP monitoring • 12-lead • $EtCO_2$ when indicated • early notification

Overview

- Rapidly distinguish an isolated mild skin reaction from anaphylaxis involving the airway, breathing, circulation, or multiple body systems.
- Epinephrine is the first-line treatment for anaphylaxis and must not be delayed for vascular access, monitoring, antihistamines, corticosteroids, or transport preparation.
- Anaphylaxis may occur without rash. Respiratory compromise or hypotension after a likely allergen exposure is sufficient to treat immediately.

Provider-Level Actions

Give epinephrine first, support airway and perfusion, and reassess continuously.

EMT

- Remove or stop the suspected trigger when safe, perform the primary assessment, position the patient appropriately, and provide oxygen, suction, or BVM ventilation when indicated.
- For anaphylaxis, give epinephrine 1 mg/mL 0.5 mg IM in the lateral thigh; a 0.3 mg autoinjector is acceptable when that is the available device. Repeat every five minutes while significant symptoms persist.
- Give albuterol 2.5 mg by nebulizer for persistent bronchospasm after epinephrine; repeat every 5-15 minutes as needed to a maximum of three treatments.
- Apply available monitoring, request ALS early, initiate transport, and do not delay epinephrine for any other treatment.

AEMT

- Perform all EMT actions and establish IV/IO access without delaying IM epinephrine or transport.

- For hypotension or poor perfusion, give isotonic crystalloid in 500 mL IV/IO aliquots with reassessment, maximum 2 liters. Use 250 mL aliquots when CHF, renal failure, liver failure, pulmonary edema, or volume overload is present or suspected.
- Continue IM epinephrine every five minutes for ongoing anaphylaxis and notify the paramedic immediately when shock, airway swelling, or refractory symptoms persist.

Paramedic

- Perform all EMT and AEMT actions; establish an early difficult-airway plan for progressive tongue, pharyngeal, or laryngeal swelling and use waveform capnography when ventilatory support or an advanced airway is required.
- For isolated or persistent skin symptoms, give diphenhydramine 25-50 mg slow IV/IO or deep IM only when mental status and blood pressure are adequate.
- For refractory shock after repeated IM epinephrine and appropriate fluid, begin an epinephrine infusion at 2-10 mcg/min and titrate to SBP at least 90 mmHg or MAP at least 65 mmHg. Medical Control is required.
- For a beta-blocked patient with refractory shock or bradycardia, give glucagon 1-2 mg IV/IO over five minutes with Medical Control and prepare for vomiting.
- For suspected ACE-inhibitor or hereditary bradykinin-mediated angioedema, consider tranexamic acid 1 g IV/IO over ten minutes with Medical Control while prioritizing airway management and transport.

Indications

- Acute allergic symptoms after a suspected exposure, including urticaria, pruritus, flushing, angioedema, wheezing, stridor, dyspnea, gastrointestinal symptoms, hypotension, collapse, or shock.
- Suspected anaphylaxis even when no rash or skin finding is present.
- Suspected ACE-inhibitor, hereditary, or other bradykinin-mediated angioedema.

Contraindications

- There is no absolute contraindication to IM epinephrine when anaphylaxis is present or strongly suspected.
- Do not give diphenhydramine to a patient with hypotension, poor perfusion, or decreased mental status.

Assessment | Severity classification

- Mild allergic reaction: isolated hives, itching, flushing, or localized swelling with a normal airway, normal breathing, and normal perfusion.
- Anaphylaxis: respiratory compromise, hypotension or poor perfusion, or acute involvement of two or more systems including skin/mucosa, respiratory, cardiovascular, or gastrointestinal symptoms.
- Acute hypotension, bronchospasm, or laryngeal involvement after a known or highly probable allergen exposure is anaphylaxis even without skin findings.

Assessment | Focused assessment

- Identify the suspected trigger, route and time of exposure, symptom onset and progression, prior anaphylaxis, epinephrine given before EMS arrival, prescribed autoinjector, asthma, mast-cell disorder, and previous biphasic reaction.
- Review beta-blocker and ACE-inhibitor use and ask about hereditary or recurrent angioedema.
- Assess voice change, drooling, dysphagia, tongue or facial swelling, stridor, wheezing, work of breathing, oxygenation, skin findings, vomiting or abdominal pain, pulse, blood pressure, perfusion, and mental status.
- Reassess airway swelling, lung sounds, blood pressure, perfusion, mental status, and response after every epinephrine dose and fluid aliquot.

Treatment steps

- Stop or remove the triggering exposure when safe, but do not delay epinephrine or transport to remove an embedded object or perform decontamination that can occur later.
- Position most patients supine when tolerated. Use a position of comfort for severe respiratory distress, left-lateral positioning during pregnancy, and the recovery position for an unconscious patient with adequate breathing.
- Give epinephrine 1 mg/mL 0.5 mg IM in the lateral thigh immediately for anaphylaxis. A 0.3 mg autoinjector is acceptable when it is the available device.
- Repeat IM epinephrine every five minutes while significant airway, respiratory, cardiovascular, or progressive multisystem symptoms persist.
- Support airway and ventilation and give albuterol 2.5 mg by nebulizer for persistent bronchospasm after epinephrine; repeat every 5-15 minutes as needed to a maximum of three treatments.
- For hypotension or poor perfusion, give isotonic crystalloid in 500 mL IV/IO aliquots with reassessment, maximum 2 liters. Use 250 mL aliquots for patients at risk of fluid overload.
- Diphenhydramine is limited to isolated or persistent skin symptoms and must never replace or delay epinephrine. Routine corticosteroids are not part of the acute anaphylaxis pathway.
- For refractory anaphylactic shock, continue IM epinephrine and begin a Medical-Control-authorized epinephrine infusion rather than giving an IV epinephrine bolus.
- Transport every patient who receives epinephrine or has airway, respiratory, cardiovascular, progressive, or multisystem symptoms with continuous reassessment and early destination notification.

Medications

Epinephrine 1 mg/mL - IM

0.5 mg IM lateral thigh; 0.3 mg autoinjector acceptable; repeat every 5 minutes

- EMT, AEMT, and Paramedic standing order for anaphylaxis.
- There is no absolute contraindication when anaphylaxis is present or strongly suspected.
- Do not delay administration for vascular access, cardiac monitoring, antihistamine, corticosteroid, or transport preparation.

Albuterol

2.5 mg nebulized; repeat every 5-15 minutes as needed, maximum 3 treatments

- For persistent bronchospasm after epinephrine.
- Albuterol does not treat upper-airway edema, hypotension, or shock and is not a substitute for repeated epinephrine.

Isotonic crystalloid

500 mL IV/IO aliquots; repeat with reassessment; maximum 2 L

- For hypotension or poor perfusion only.
- Use 250 mL aliquots for CHF, renal failure, liver failure, pulmonary edema, or volume-overload risk.

Diphenhydramine

25-50 mg slow IV/IO or deep IM

- Paramedic only for isolated or persistent skin symptoms; EMT/AEMT administration is not enabled without documented specific Tennessee authority.
- Do not administer for hypotension, poor perfusion, or decreased mental status.
- Adjunct only; never delay or replace epinephrine.

Epinephrine infusion

2-10 mcg/min IV/IO; titrate to SBP ≥90 mmHg or MAP ≥65 mmHg

- Paramedic with Medical Control for refractory shock after repeated IM epinephrine and appropriate fluid.
- Use an infusion pump and continuous cardiac, blood-pressure, and SpO_2 monitoring.
- Do not give an IV epinephrine bolus under this protocol.

Glucagon

1-2 mg IV/IO over 5 minutes

- Paramedic with Medical Control for refractory shock or bradycardia in a beta-blocked patient.
- Prepare for vomiting and protect the airway.

Tranexamic acid

1 g IV/IO over 10 minutes

- Paramedic with Medical Control for suspected ACE-inhibitor or hereditary bradykinin-mediated angioedema.
- Evidence is limited; do not delay difficult-airway preparation or transport.

Warnings / precautions

- Anaphylaxis may occur without rash or hives.
- Do not delay initial or repeat IM epinephrine for IV/IO access, monitoring, albuterol, diphenhydramine, corticosteroids, or transport preparation.
- Do not give an IV epinephrine bolus for anaphylaxis under this protocol; dosing errors can cause severe hypertension, myocardial ischemia, and lethal dysrhythmia.
- Do not allow a hypotensive or poorly perfused patient to stand or walk because abrupt cardiovascular collapse may occur.
- ACE-inhibitor and hereditary angioedema may progress rapidly and may respond poorly to epinephrine, antihistamines, and corticosteroids.
- Routine corticosteroids are not recommended for acute anaphylaxis and have not been shown to reliably prevent a biphasic reaction.

Clinical pearls

- The shorter the interval from exposure to symptom onset, the greater the concern for rapid progression.
- A normal skin examination does not exclude anaphylaxis; prioritize airway, breathing, circulation, perfusion, and gastrointestinal findings.
- Albuterol treats lower-airway bronchospasm but does not treat laryngeal edema or shock.
- Obtain a 12-lead ECG and continuous cardiac monitoring in moderate or severe reactions without delaying epinephrine.
- Use continuous SpO_2 and serial blood pressure monitoring; add $EtCO_2$ when ventilation is supported or airway compromise is present.
- Document the suspected trigger, exact symptom-onset time, every epinephrine dose and route, response, recurrence, and all treatment given before EMS arrival.

Special populations | Pregnancy

- Use left-lateral positioning when feasible and treat anaphylaxis with the same prompt IM epinephrine dosing.
- Maternal oxygenation and perfusion are the immediate priorities for fetal resuscitation.

Special populations | Beta-blocker therapy

- Expect possible resistance to epinephrine and increased risk of refractory hypotension or bradycardia.
- Glucagon may be used with Medical Control; give slowly and prepare for vomiting.

Special populations | ACE-inhibitor or hereditary angioedema

- Prioritize early difficult-airway preparation and rapid transport.
- Give epinephrine when anaphylaxis cannot be excluded or airway compromise is developing, while recognizing that nonhistaminergic angioedema may respond poorly.
- TXA is a Medical-Control-directed formulary option with limited supporting evidence.

Related pathways / contacts

AR-01 Adult Airway - Open for progressive airway swelling, ventilatory failure, or advanced-airway management.

AR-04 COPD / Asthma - Open for persistent bronchospasm or a concurrent asthma/COPD exacerbation.

AM-05 Hypotension / Shock - Open for persistent hypotension or another shock etiology after anaphylaxis treatment.

Source and references

Golden DBK, et al. Anaphylaxis: A 2023 Practice Parameter Update. Annals of Allergy, Asthma & Immunology. 2024;132(2):124-176. | Cardona V, et al. World Allergy Organization Anaphylaxis Guidance 2020. World Allergy Organization Journal. 2020;13:100472. | Dodd A, et al. Evidence Update for the Treatment of Anaphylaxis. Resuscitation. 2021;163:86-96. | Centers for Disease Control and Prevention. Emergency Management of Anaphylaxis in Adults, reviewed July 25, 2024. | North Carolina College of Emergency Physicians. AM-01 Allergic Reaction / Anaphylaxis Protocol, revised September 1, 2025. | Claiborne County EMS formulary and medical-director decisions, October 2026.

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